

YASH RAWAL

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EXPERIENCE

Bigscal Technologies Pvt. Ltd.

Surat, India

AI/ML Engineer

Oct. 2024 – Present

- Deployed enterprise agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ cloud tenants.
- Architected multi-agent routing with **LangGraph & SAP Joule 2.0**, integrating hybrid tool discovery across 650+ MCP tools.
- Re-architected multi-agent execution** with capability contracts, delivering **3× throughput, 70% token cut, and \$120K+ savings**.
- Fine-tuned and served **Qwen2.5-7B & E5-large-v2** models on NVIDIA GPUs using PyTorch, Unsloth, and contrastive learning via vLLM.
- Implemented 4-level hierarchical RAG pipeline in Python indexing 10,000+ technical documents using PGVector (**90%+ Recall@10**).
- Built Prompt Lab evaluation control plane to monitor model outputs, collecting reasoning trajectories for **RLHF benchmarking**.

Bigscal Technologies Pvt. Ltd.

Surat, India

AI/ML Engineer Intern

Apr. 2024 – Sep. 2024

- Engineered clinical NLP summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3 and FastAPI endpoints.
- Built real-time WebSocket audio streaming for Indic ASR with **NVIDIA NeMo (RNN-T and CTC decoding)** and torchaudio DSP.

PROJECTS

vLLM High-Performance Inference Engine | Python, PyTorch, Pydantic, AsyncIO, PyTest

Open Source

- Merged **PR #55475** fixing structural XML tool parser drops on `/v1/chat/completions` and engineered torchcodec multimodal guards.

Job Architecture Generator (Qwen2.5-7B SFT) | Python, PyTorch, Hugging Face, Unsloth, vLLM

LLM Training

- Fine-tuned **Qwen2.5-7B** on 10K+ domain examples using Unsloth on NVIDIA GPUs, deploying OpenAI-compatible vLLM endpoints.

E5-large-v2 Job & Skills Embedding Fine-Tuning | Python, PyTorch, Triplet Loss, Transformers

Model Training

- Fine-tuned **E5-large-v2** (335M params) on labor data with triplet loss, raising semantic retrieval relevance by **18%** over OpenAI.

Modular Agent Harness & Conversational Copilot | Python, FastAPI, PostgreSQL, WebSockets, pgvector

Project

- Architected modular Python agentic framework implementing formal ReAct loop with step budgets and pgvector tool routing.

Hierarchical Document Intelligence & RAG Platform | Python, PostgreSQL, pgvector, GROBID

Project

- Engineered immutable 4-level hierarchical RAG ingestion (paragraphs, pages, headings) using GROBID and PGVector retrieval.

TECHNICAL SKILLS

Languages & Backend: Python, SQL, FastAPI, PostgreSQL, Oracle, REST APIs, WebSockets, AsyncIO, Linux

AI & ML Engineering: Large Language Models (LLMs), Agentic AI, MCP, Tool RAG, Model Fine-Tuning (SFT/LoRA), vLLM Serving, RLHF

Frameworks & Libraries: LangChain, LangGraph, vLLM, PyTorch, Hugging Face Transformers, SAP Joule 2.0, NVIDIA NeMo, LlamaIndex

Cloud & DevOps: Azure, AWS, Docker, Kubernetes, CI/CD (GitHub Actions), Redis, PGVector, PyTest, LLM-as-a-Judge

EDUCATION & CERTIFICATIONS

Parul University

Gujarat, India

Master of Computer Applications (Artificial Intelligence & Machine Learning)

2022 – 2024

Certifications: Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft Certified: Azure Fundamentals (AZ-900)